CS5234 Quiz 1 · Mock Paper

Solutions & marking

Format (as announced): 50 minutes · open book · Lectures 1–5 · 4 MCQ (1 pt each) + 2 essay questions (9 + 7 pts) · a blank essay question earns 1 mark.

Part I · Multiple choice (4 × 1 pt)

MCQ 1 (L1)

Let $X_1, \ldots, X_k$ be *pairwise* independent Bernoulli(p) variables and $X = \sum_i X_i$. Which statement is always true?

(A) $\mathrm{Var}(X) = kp(1-p)$

(B) $\Pr(|X - kp| \geq \delta kp) \leq 2e^{-\delta^2 kp/3}$ for $0 < \delta \leq 1$

(C) $\mathbb{E}[X^2] = k^2p^2$

(D) X is independent of X_1

Solution. (A). Pairwise independence makes every covariance 0, so the variances add: $\mathrm{Var}X = \sum \mathrm{Var}X_i = kp(1-p)$. (B) is Chernoff, which needs *full* independence. (C) forgets the variance: $\mathbb{E}X^2 = \mathrm{Var}X + (\mathbb{E}X)^2 = kp(1-p) + k^2p^2$. (D) is false because X contains X_1.

MCQ 2 (L2)

An estimator is within εA of the true value A with probability 0.6. How many independent repetitions are needed so that the *median* is within εA with probability $1 - \delta$?

(A) $\Theta(1/\delta)$

(B) $\Theta(\log(1/\delta))$

(C) impossible, because $0.6 < 2/3$

(D) $\Theta(1/\delta^2)$

Solution. (B). The median trick only needs single-run success strictly above $1/2$; the median fails only if at least half of the runs fail on the same side, and Chernoff makes that $e^{-\Omega(k)}$ (the constant in Ω depends on $0.6 - 0.5$). The number $2/3$ is just a convention for "constant success", not a threshold.

MCQ 3 (L3)

Which statement about Yao's principle is correct?

(A) $D_\mu(f) \geq R(f)$ for every input distribution μ

(B) To prove $R(f) \geq q$ it suffices to exhibit one distribution μ with $D_\mu(f) \geq q$

(C) It only applies to problems with a 0/1 answer

(D) The hard distribution must be uniform over all inputs

Solution. (B). Yao gives $D_\mu(f) \leq R(f)$ for every μ, so (A) has the inequality reversed and (B) follows. The averaging proof never uses that the output is binary, so it works for search problems too (e.g. finding the maximum). For OR the hard distribution is $\frac{1}{2}$ all-zero $+\frac{1}{2}$ one random 1, which is far from uniform, so (D) is false.

MCQ 4 (L4)

For testing whether an array of n distinct numbers is sorted (ε-far means more than εn entries must change), which tester is correct with $O(\log n/\varepsilon)$ queries?

(A) Pick $O(1/\varepsilon)$ random i and check $a_i \leq a_{i+1}$

(B) Pick $O(1/\varepsilon)$ random indices, sort them, and check the values are increasing

(C) Pick $O(1/\varepsilon)$ random indices and run a binary search for each value

(D) None: any tester needs $\Omega(n)$ queries

Solution. **(C).** (A) fails on $7, 8, \ldots, 12, 1, \ldots, 6$: $\frac{1}{2}$-far but only one adjacent inversion. (B) fails on $2, 1, 4, 3, \ldots$: it needs both elements of a swapped pair, which takes about $\sqrt{n}$ samples. (D) is true only for *exact* sortedness; testing is sublinear. (C) works: indices whose binary search succeeds are sorted among themselves, so an ε-far array has at least εn failing indices.

Part II · Essay questions

Essay 1 (9 pts) · Counting equal pairs

You have query access to an array $a_1, \ldots, a_n$. Let $N = \binom{n}{2}$ and let P be the number of pairs $i < j$ with $a_i = a_j$.

(a) [2] Using two queries, give an unbiased estimator of P.

Solution. Pick a pair $\{i, j\}$ uniformly among the N pairs, query a_i, a_j, and output $X = N \cdot \mathbf{1}[a_i = a_j]$. Then $\mathbb{E}X = N \cdot \frac{P}{N} = P$. *Marking: 1 pt estimator, 1 pt expectation.*

(b) [3] Estimate P within additive error εN with probability $1 - \delta$ using $O(\varepsilon^{-2} \log(1/\delta))$ queries.

Solution. Take k independent pairs; let $Y_t = \mathbf{1}[\text{pair } t \text{ is equal}] \in \{0, 1\}$, independent with mean $\rho = P/N$, and output $\hat{P} = N \cdot \frac{1}{k} \sum_t Y_t$. Error $\le \varepsilon N \Leftrightarrow |\bar{Y} - \rho| \le \varepsilon$. Hoeffding: $\Pr[|\bar{Y} - \rho| > \varepsilon] \le 2e^{-2k\varepsilon^2} \le \delta$ for $k = \lceil \ln(2/\delta)/(2\varepsilon^2) \rceil$; queries $2k = O(\varepsilon^{-2} \log(1/\delta))$. (Equivalently: Chernoff as in the L2 counting analysis gives $k = O(1/\varepsilon^2)$ for probability $2/3$, then the median trick with $O(\log(1/\delta))$ repetitions.) *Marking: 1 pt independence stated, 1 pt correct inequality, 1 pt final k and query count.*

(c) [4] Now suppose we must decide exactly whether $P = 0$ or $P \ge 1$, with success probability $\ge 2/3$. Prove that $\Omega(n)$ queries are necessary. *Hint: let μ_0 be $a_i = i$ for all i; for μ_1, copy a_1 into one random position.*

Solution. *Hard distributions (1 pt)*: μ_0: $a_i = i$, so $P = 0$. μ_1: pick J uniform in $\{2, \ldots, n\}$ and set $a_J = 1$, other $a_i = i$, so $P = 1$. Draw $I \in \{0, 1\}$ uniformly and $a \sim \mu_I$. *Transcripts (2 pts)*: fix a deterministic algorithm with q queries. As long as every answer is $a_i = i$, it follows one fixed path querying fixed positions $p_1, \ldots, p_q$; let τ_0 be that transcript. Under μ_0 the transcript is always τ_0; under μ_1 it differs from τ_0 only if $J \in \{p_1, \ldots, p_q\}$, which has probability $\le \frac{q}{n-1}$. Hence $\Delta_{\mathrm{TV}}(\nu_0, \nu_1) = 1 - \nu_1(\tau_0) \le \frac{q}{n-1}$. *Conclusion (1 pt)*: the success probability is at most $\frac{1}{2} + \frac{1}{2}\Delta_{\mathrm{TV}} \le \frac{1}{2} + \frac{q}{2(n-1)}$, so reaching $\frac{2}{3}$ needs $q \ge \frac{n-1}{3}$. Thus $D_\mu \ge \frac{n-1}{3}$ and by Yao $R = \Omega(n)$. *Remark*: this does not contradict (b), because an additive error of $\varepsilon N = \Theta(\varepsilon n^2)$ cannot tell $P = 0$ from $P = 1$.

Essay 2 (7 pts) · Adjacent repeats in a stream

A stream $x_1, \ldots, x_n \in [m]$ arrives one element at a time; n is not known in advance. Let $R = |\{i \le n - 1 : x_i = x_{i+1}\}|$ and $\rho = R/(n-1)$. We want $\hat{\rho}$ with $|\hat{\rho} - \rho| \le \varepsilon$ with probability at least $2/3$.

(a) [3] Design a streaming algorithm. *Hint: sample a uniformly random position $I \in [n-1]$ with reservoir sampling, but you also need x_{I+1}.*

Solution. Keep the previous element in a one-word buffer. Run k independent copies of the following. When x_i arrives ($i \ge 2$), position $i - 1$ becomes eligible: with probability $\frac{1}{i-1}$, replace the stored pair by (x_{i-1}, x_i) (taking x_{i-1} from the buffer). By the L5 reservoir argument over positions $1, \ldots, n-1$, at the end each copy holds (x_I, x_{I+1}) for a uniform $I \in [n-1]$. Output $\hat{\rho} = \frac{1}{k} \sum_t \mathbf{1}[\text{copy } t \text{ holds an equal pair}]$. *Marking: 1 pt buffer idea, 1 pt correct replacement probability, 1 pt output.*

(b) [2] How large must k be? Justify.

Solution. The indicators Z_t are independent (independent copies) and each has mean $\Pr[x_I = x_{I+1}] = \rho$. Hoeffding: $\Pr[|\hat{\rho} - \rho| > \varepsilon] \le 2e^{-2k\varepsilon^2} \le \frac{1}{3}$ for $k = \lceil \ln 6/(2\varepsilon^2) \rceil = O(1/\varepsilon^2)$. (Chebyshev also works: $\mathrm{Var}\hat{\rho} \le \frac{1}{4k}$, so $k = \frac{3}{4\varepsilon^2}$.)

(c) [1] What is the space complexity?

Solution. k stored pairs ($2 \log m$ bits each), one buffer word ($\log m$) and a position counter ($\log n$): $O(\varepsilon^{-2} \log m + \log n)$ bits.

(d) [1] Why does it *not* work to simply store the first k adjacent pairs of the stream?

Solution. The first k positions are not a random sample. An adversarial stream can put all its repeats after position k (e.g. $1, 2, \ldots, n/2$ followed by $n/2$ copies of the same value): then $\rho \approx \frac{1}{2}$ but the first k pairs show no repeats, so the estimate is 0. Concentration bounds need each sample to be uniform over all positions.